AWS GLUE CRAWLER

APARTADO A

1.- Desde Aws CLI explora el contenido del bucket s3://noaa-ghcn-pds/csv/.

Utilizando el comando ls podemos ver que contiene dos carpetas, y ambas de ellas contienen múltiples archivos .csv

```
PS C:\WINDOWS\System32> aws s3 ls s3://noaa-ghcn-pds/csv/
                           PRE by_station/
                           PRE by_year/
```

```
PS C:\WINDOWS\System32> aws s3 ls s3://noaa-ghcn-pds/csv/by_station/
```

```
2025-02-09 08:54:27     192697 WA010050800.csv
2025-02-09 08:54:27     191686 WA010063530.csv
2025-02-09 08:54:27     158415 WA010066250.csv
2025-02-09 08:54:27     167676 WA010066890.csv
2025-02-09 08:54:27     170768 WA010082900.csv
2025-02-09 08:54:28     177587 WA010085780.csv
2025-02-09 08:54:27     195958 WA010086600.csv
2025-02-09 08:54:27     166587 WA010094670.csv
2025-02-09 08:54:27     192891 WA010095930.csv
2025-02-09 08:54:27     196272 WA010096390.csv
2025-02-09 08:54:27     196151 WA010098500.csv
2025-02-09 08:54:27     190440 WA010098520.csv
2025-02-09 08:54:28     194501 WA010101180.csv
2025-02-09 08:54:29    2498147 WA010101860.csv
2025-02-09 08:54:27     129119 WA010107010.csv
2025-02-09 08:54:28     191378 WA010111300.csv
2025-02-09 08:54:27     129194 WA010117880.csv
2025-02-09 08:54:27     188571 WA010150350.csv
2025-02-09 08:54:27     165087 WA010472480.csv
2025-02-09 08:54:28     482604 WA010517310.csv
2025-02-09 08:54:27     216814 WA010518000.csv
2025-02-09 08:54:27     195641 WA010526410.csv
2025-02-09 08:54:27     196019 WA010542310.csv
2025-02-09 08:54:28     195401 WA010543290.csv
2025-02-09 08:54:27     195356 WA010543500.csv
2025-02-09 08:54:28     858317 WA010553740.csv
2025-02-09 08:54:27     192050 WA010554470.csv
2025-02-09 08:54:27     193355 WA010558090.csv
2025-02-09 08:54:27     197081 WA010574650.csv
2025-02-09 08:54:27     195993 WA010576160.csv
```

```
PS C:\WINDOWS\System32> aws s3 ls s3://noaa-ghcn-pds/csv/by_year/
```

```
PS C:\Users\Mañana> aws s3 ls s3://noaa-ghcn-pds/csv/by_year/
2025-01-14 13:42:27   11608918 1750.csv
2026-01-12 18:29:55      25333 1763.csv
2026-01-12 18:29:56      25340 1764.csv
2026-01-12 18:29:53      25328 1765.csv
2026-01-12 18:29:57      25378 1766.csv
2026-01-12 18:30:01      25374 1767.csv
2026-01-12 18:30:02      25419 1768.csv
2026-01-12 18:30:04      25310 1769.csv
2026-01-12 18:30:04      25316 1770.csv
2026-01-12 18:30:07      25305 1771.csv
2026-01-12 18:30:02      25449 1772.csv
2026-01-12 18:29:54      25305 1773.csv
2026-01-12 18:29:58      25354 1774.csv
2026-01-12 18:30:04      50554 1775.csv
2026-01-12 18:30:04      50705 1776.csv
2026-01-12 18:30:01      50608 1777.csv
2026-01-12 18:30:07      50348 1778.csv
2026-01-12 18:29:57      49631 1779.csv
2026-01-12 18:30:06      50536 1780.csv
2026-01-12 18:30:00      62578 1781.csv
2026-01-12 18:29:54      62494 1782.csv
2026-01-12 18:30:07      62777 1783.csv
2026-01-12 18:30:03      62986 1784.csv
2026-01-12 18:30:01      62786 1785.csv
2026-01-12 18:30:04      62764 1786.csv
2026-01-12 18:30:08      50581 1787.csv
2026-01-12 18:30:06      50633 1788.csv
2026-01-12 18:30:05      62776 1789.csv
```

2.- Descarga uno cualquiera de los archivos que contiene en cada una de sus carpetas y muestra las primeras líneas de ellos.

Utilizamos cp para descargar los archivos

```
PS C:\Users\Mañana> aws s3 cp s3://noaa-ghcn-pds/csv/by_station/WA010050800.csv ./
download: s3://noaa-ghcn-pds/csv/by_station/WA010050800.csv to .\WA010050800.csv
```

```
PS C:\Users\Mañana> aws s3 cp s3://noaa-ghcn-pds/csv/by_year/1764.csv ./
download: s3://noaa-ghcn-pds/csv/by_year/1764.csv to .\1764.csv
PS C:\Users\Mañana>
```

Y utilizamos Get-Content con –First 5 para mostrar las cinco primeras líneas

```
PS C:\Users\Mañana> Get-Content WA010050800.csv -First 5
ID,DATE,ELEMENT,DATA_VALUE,M_FLAG,Q_FLAG,S_FLAG,OBS_TIME
WA010050800,19680101,PRCP,50,,,I,
WA010050800,19680102,PRCP,0,,,I,
WA010050800,19680103,PRCP,0,,,I,
WA010050800,19680104,PRCP,0,,,I,
PS C:\Users\Mañana> Get-Content 1764.csv -First 5
ID,DATE,ELEMENT,DATA_VALUE,M_FLAG,Q_FLAG,S_FLAG,OBS_TIME
ITE00100554,17640101,TMAX,64,,,E,
ITE00100554,17640101,TMIN,44,,,E,
ITE00100554,17640102,TMAX,46,,,E,
ITE00100554,17640102,TMIN,26,,,E,
```

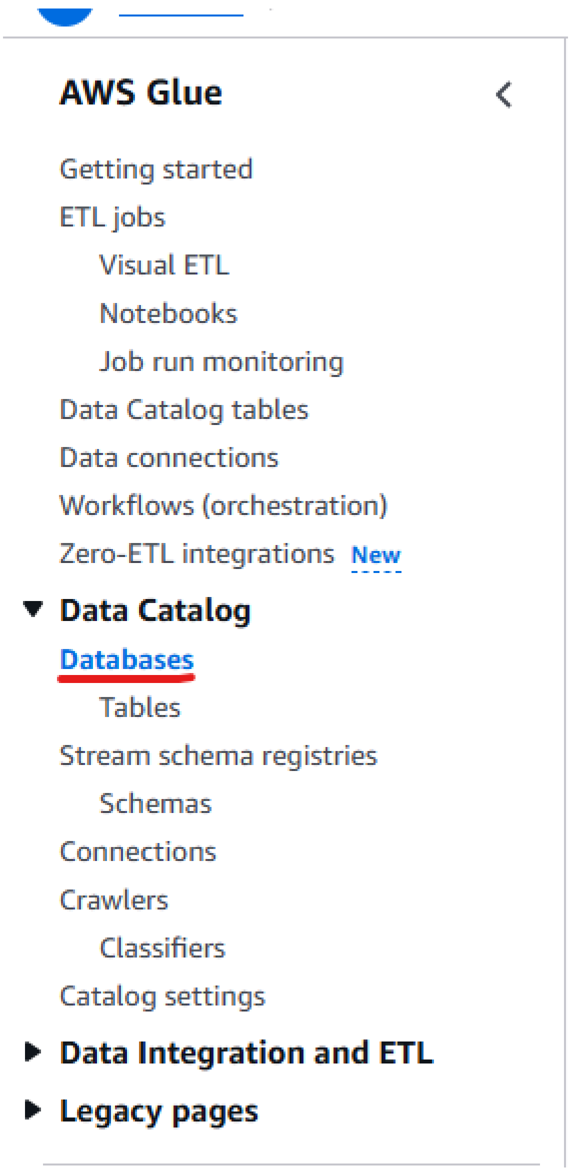
3.- ¿Qué contiene cada uno de los dos tipos de archivos?

Ambos archivos contienen los datos del clima medidos en diferentes estaciones. Los del archivo WA010050800 contiene los de dicha estación, mientras que los del archivo 1764 contiene los de ese año en específico.

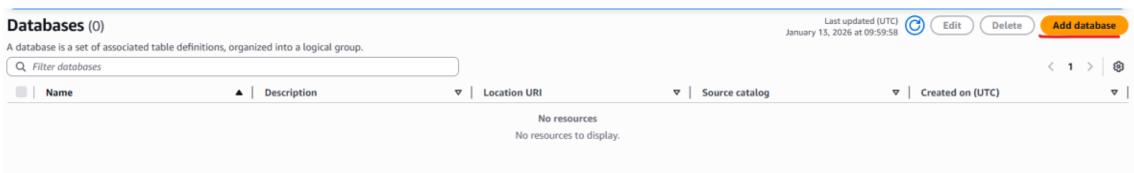
APARTADO B

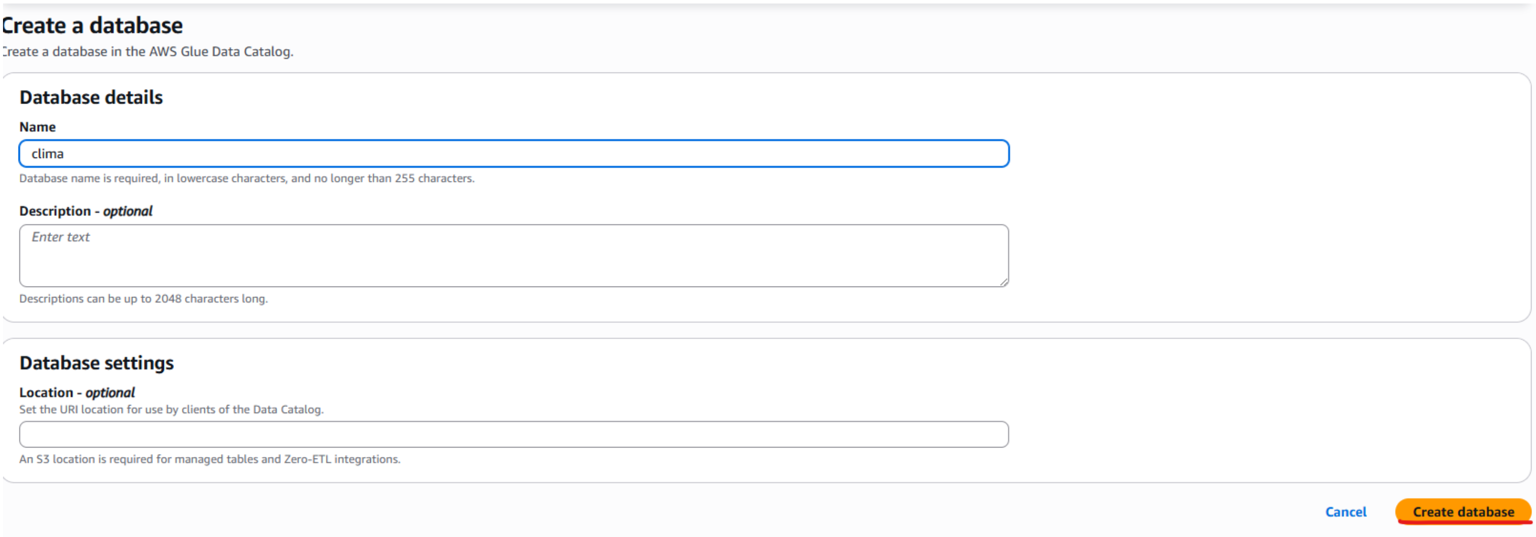
1.- Crea una base de datos en AWE GLUE llamada clima.

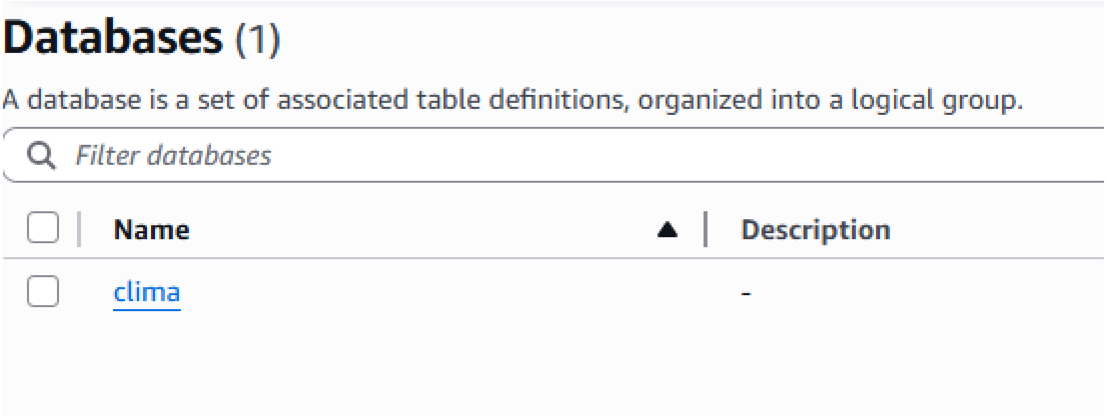
En AWS glue hacemos click en el apartado Data catalog en el menú de la izquierda y después en “Databases”



Después en Add Database

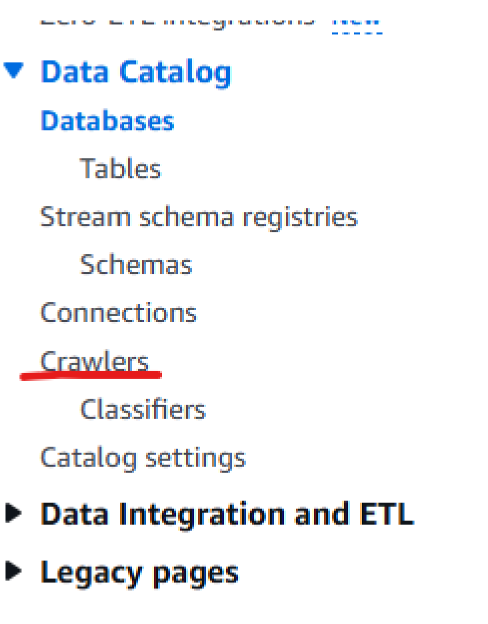




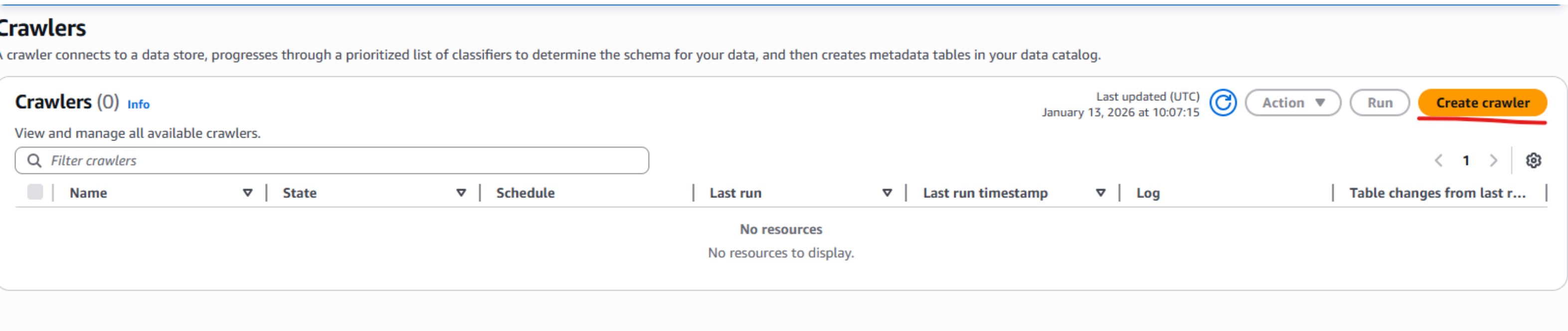


2.- Crea un Crawler AWS GLUE que nos explore el bucket del ejercicio anterior generando las tablas en la base de datos que acabas de crear.

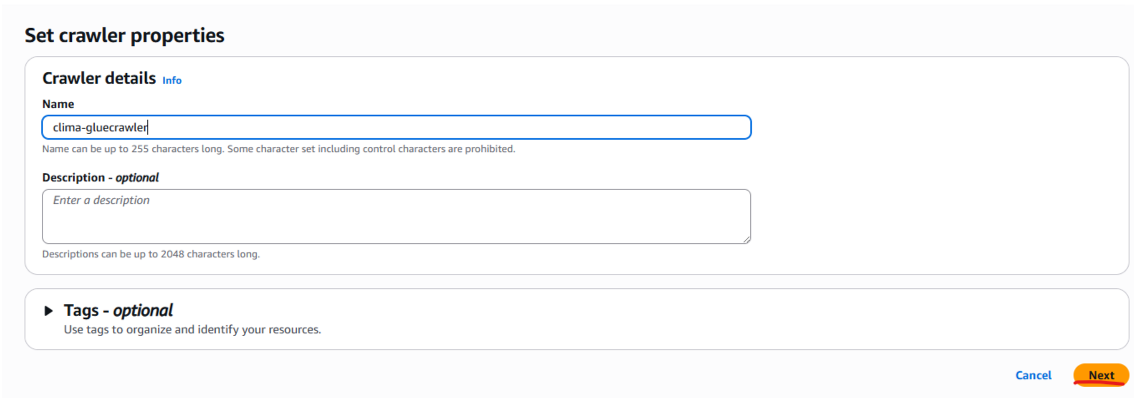
De nuevo en el menú de la izquierda en Data Catalog, hacemos click en Crawlers

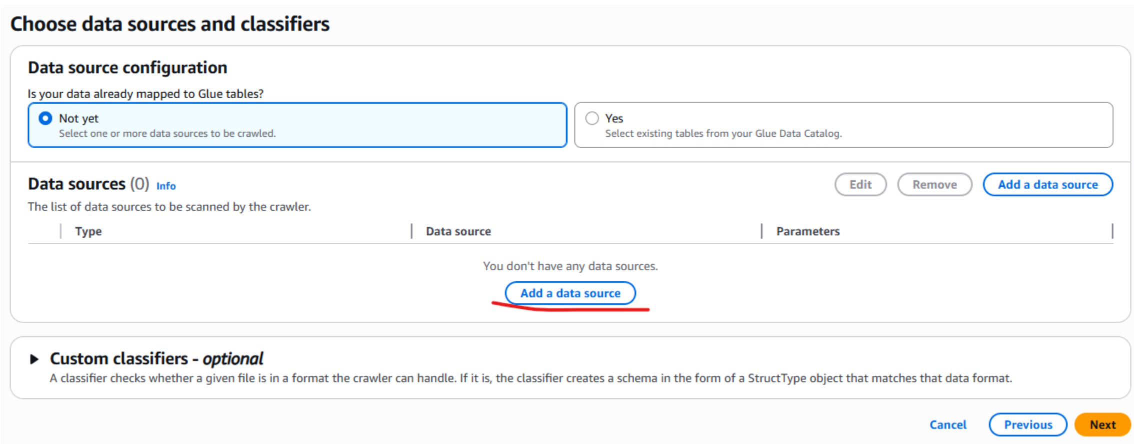


Y después a Create Crawler



Seguimos los pasos que se nos indican





Add data source

Data source

Choose the source of data to be crawled.

S3

Network connection - *optional*

Optionally include a Network connection to use with this S3 target. Note that each crawler is limited to one Network connection so any other S3 targets will also use the same connection (or none, if left blank).

Clear selection

Add new connection

Location of S3 data

- In this account
- In a different account

S3 path

Browse for or enter an existing S3 path.

s3://noaa-ghcn-pds/csv/

All folders and files contained in the S3 path are crawled. For example, type s3://MyBucket/MyFolder/ to crawl all objects in MyFolder within MyBucket.

Subsequent crawler runs

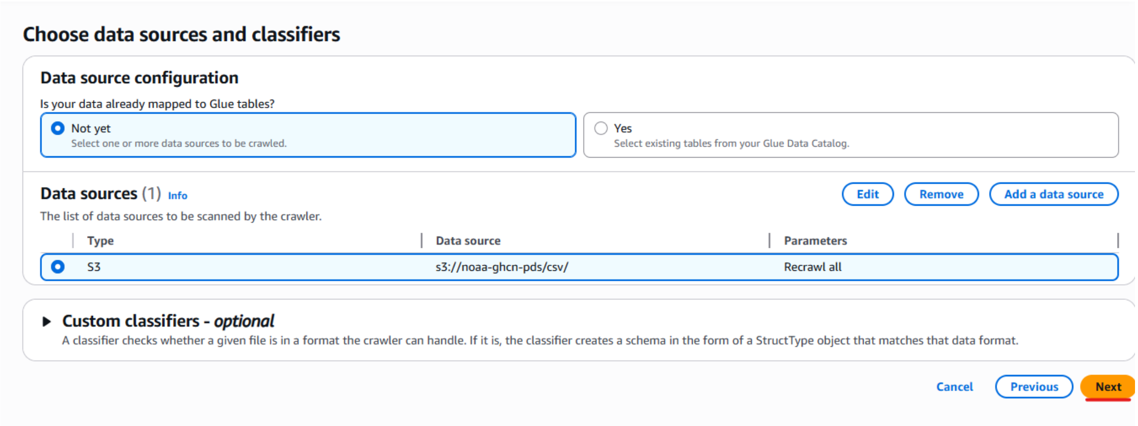
This field is a global field that affects all S3 data sources.

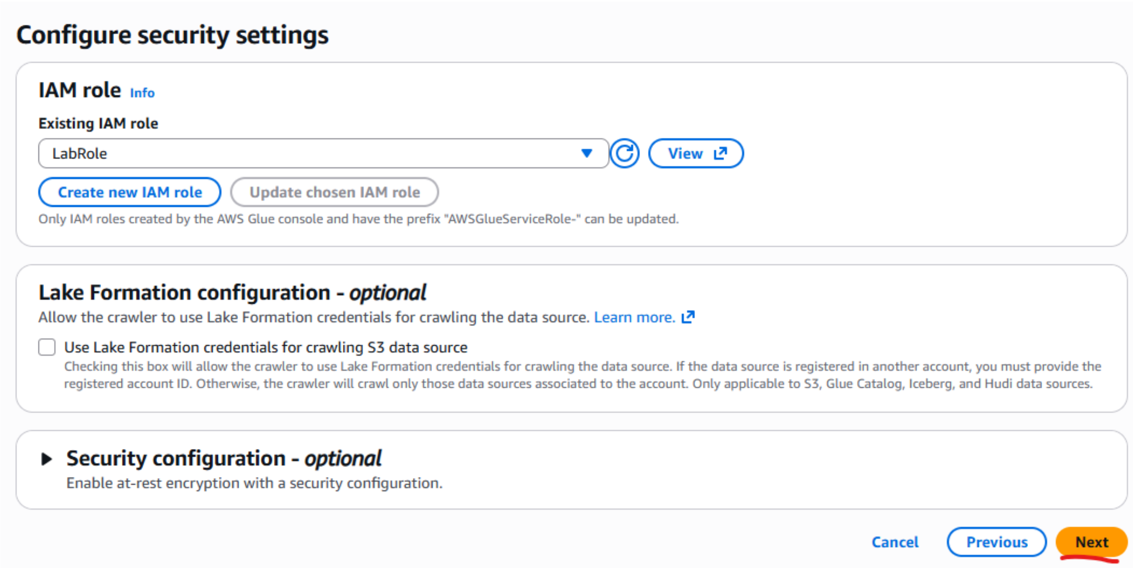
- Crawl all sub-folders
  Crawl all folders again with every subsequent crawl.
- Crawl new sub-folders only
  Only Amazon S3 folders that were added since the last crawl will be crawled. If the schemas are compatible, new partitions will be added to existing tables.
- Crawl based on events
  Rely on Amazon S3 events to control what folders to crawl.

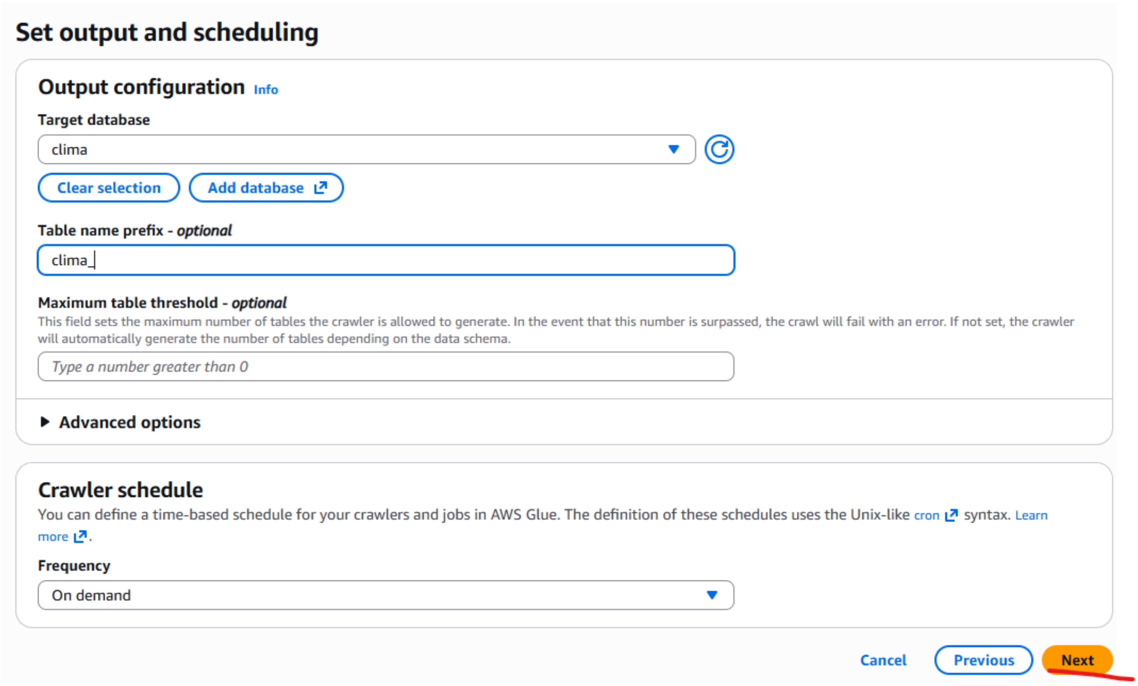
- [ ] Sample only a subset of files
- [ ] Exclude files matching pattern

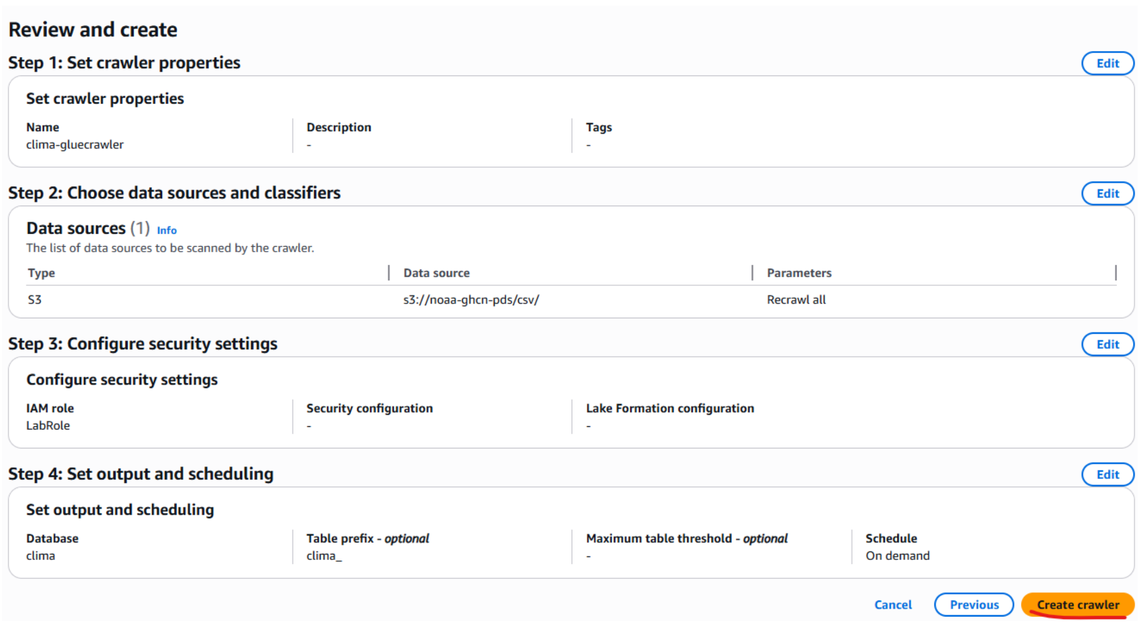
Cancel

Add an S3 data source

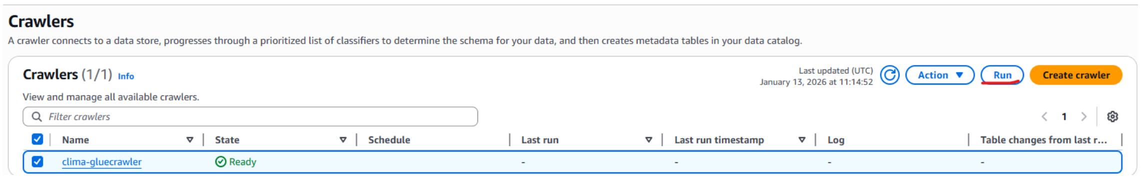

Choose data sources and classifiers
Data source configuration
Is your data already mapped to Glue tables?
Not yet
Select one or more data sources to be crawled.
Yes
Select existing tables from your Glue Data Catalog.
Data sources (1) Info
Edit
Remove
Add a data source
The list of data sources to be scanned by the crawler.
Type
Data source
Parameters
S3
s3://noaa-ghcn-pds/csv/
Recrawl all
Custom classifiers - optional
A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.
Cancel
Previous
Next


Configure security settings
IAM role Info
Existing IAM role
LabRole
View
Create new IAM role
Update chosen IAM role
Only IAM roles created by the AWS Glue console and have the prefix "AWSGlueServiceRole-" can be updated.
Lake Formation configuration - optional
Allow the crawler to use Lake Formation credentials for crawling the data source. Learn more.
Use Lake Formation credentials for crawling S3 data source
Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, you must provide the registered account ID. Otherwise, the crawler will crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.
Security configuration - optional
Enable at-rest encryption with a security configuration.
Cancel
Previous
Next


Set output and scheduling
Output configuration Info
Target database
clima
Clear selection
Add database
Table name prefix - optional
clima_
Maximum table threshold - optional
This field sets the maximum number of tables the crawler is allowed to generate. In the event that this number is surpassed, the crawl will fail with an error. If not set, the crawler will automatically generate the number of tables depending on the data schema.
Type a number greater than 0
Advanced options
Crawler schedule
You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like cron syntax. Learn more.
Frequency
On demand
Cancel
Previous
Next

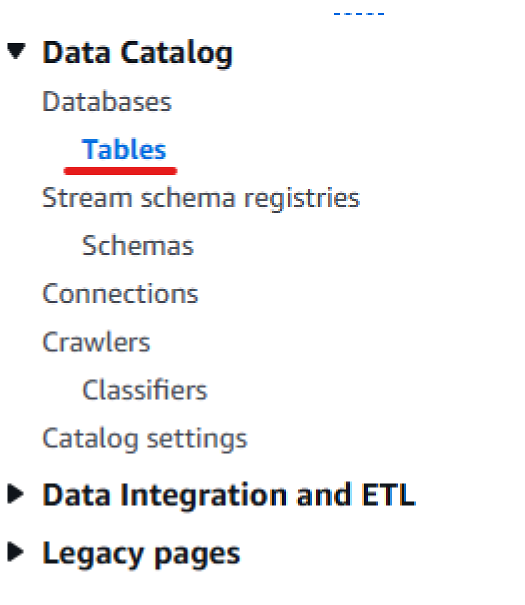


Una vez creado, volvemos a Crawlers, seleccionamos el crawler que acabamos de crear y hacemos click en Run.

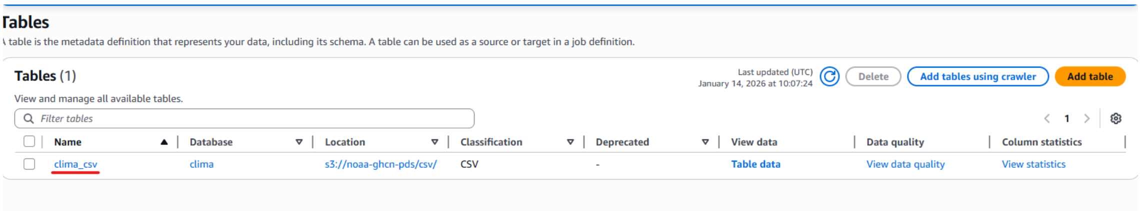


3.- Desde el apartado de Tablas de AWS GLUE, muestra la descripción del esquema de las tablas detectadas y el resumen de estadístico de sus columnas.

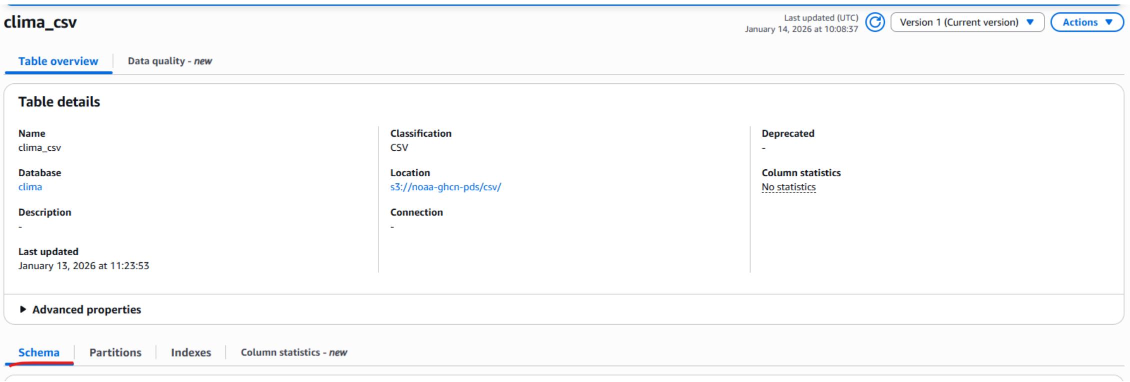
En “Data-catalog”, hacemos click en “Tablas”

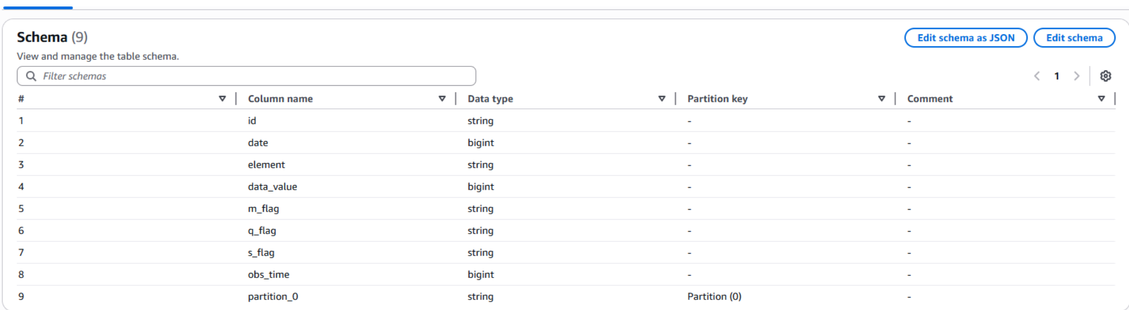


Elegimos la tabla correspondiente

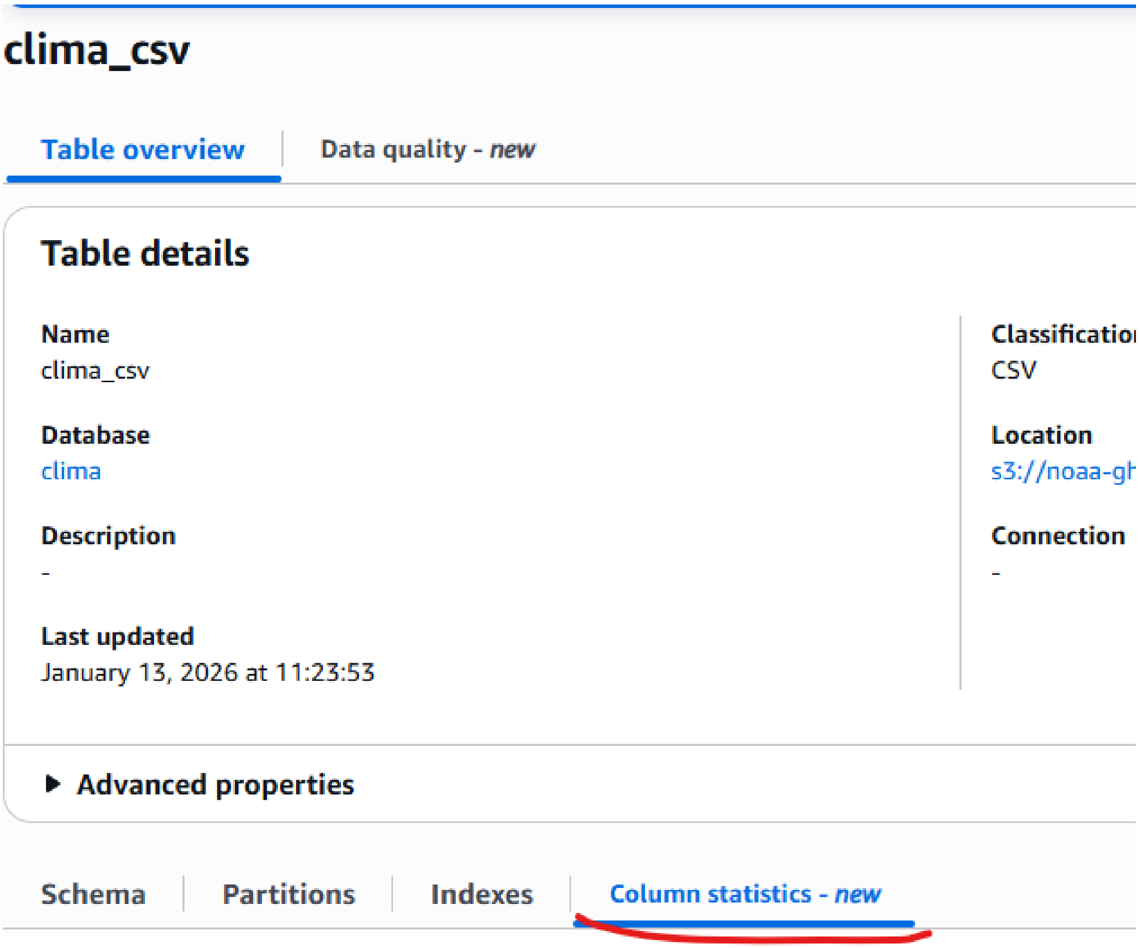


Una vez dentro, para ver el esquema de las tablas selecciona “Schema”

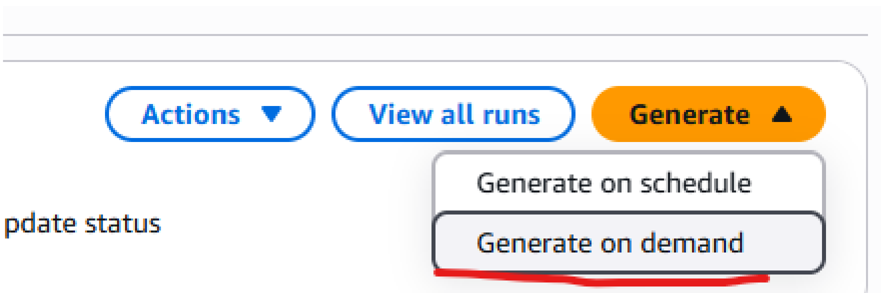




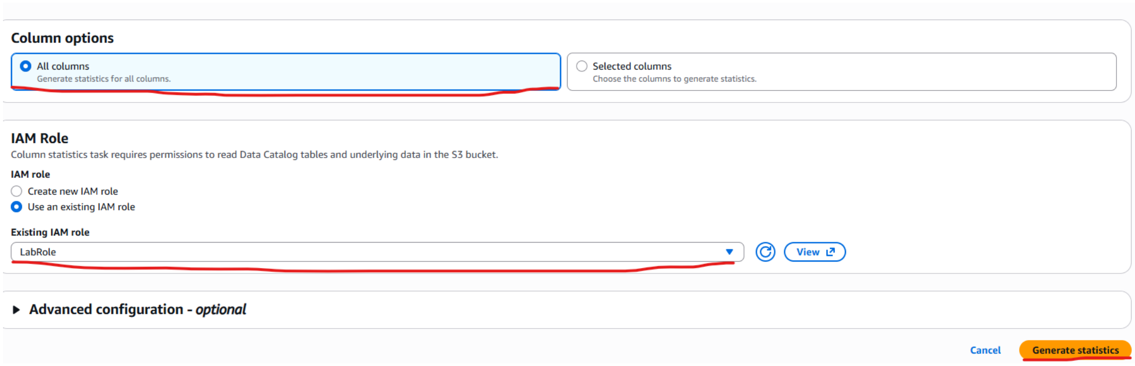
Para ver el resumen estadístico, hacemos click en “Column Statistics”



Después, hacemos click en generate y “generate on demand”



En las opciones Escogemos “All columns”, le asignamos un rol IAM (En este caso LabRole) y hacemos click en “Generate Statistics”



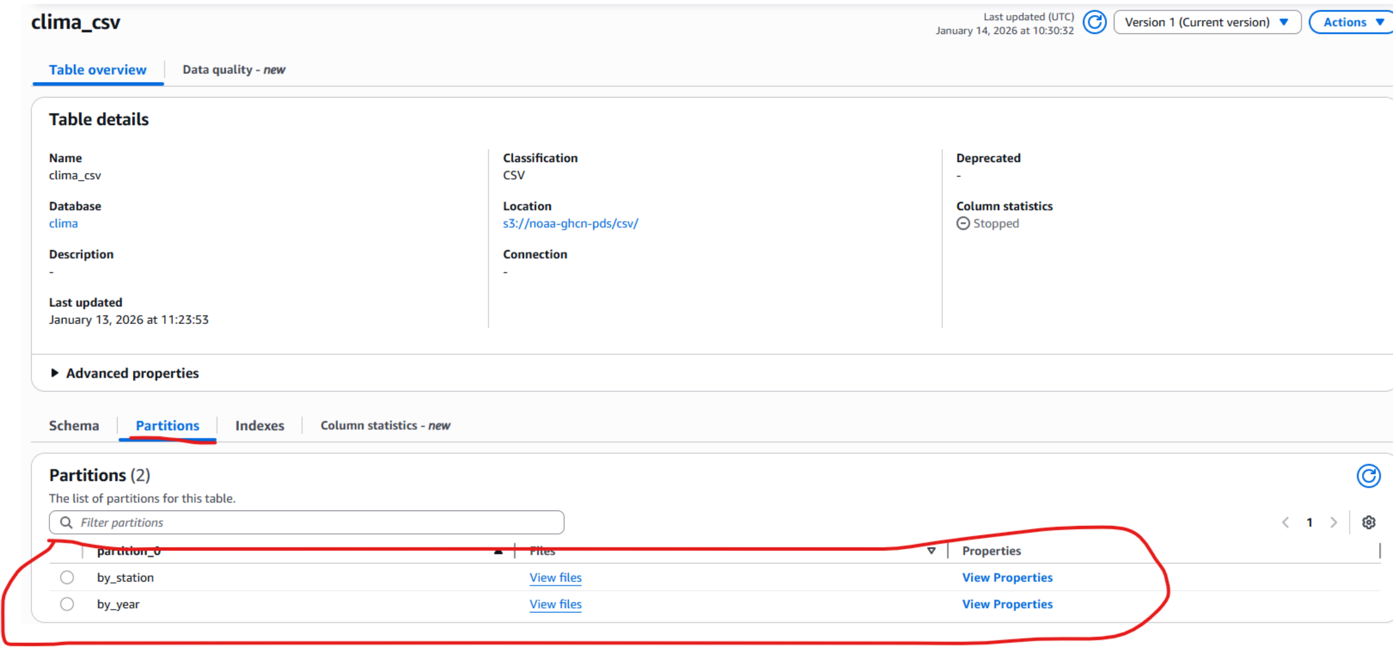
Una vez terminen de generarse, volvemos a la tabla y observamos las estadísticas en “Column Statistics”

No se han podido enseñar debido a que se bloquea a la hora de realizarlas

4.- ¿Está particionada la tabla? ¿Por qué campos?

En el apartado “Partitions” podemos ver que la tabla está particionada

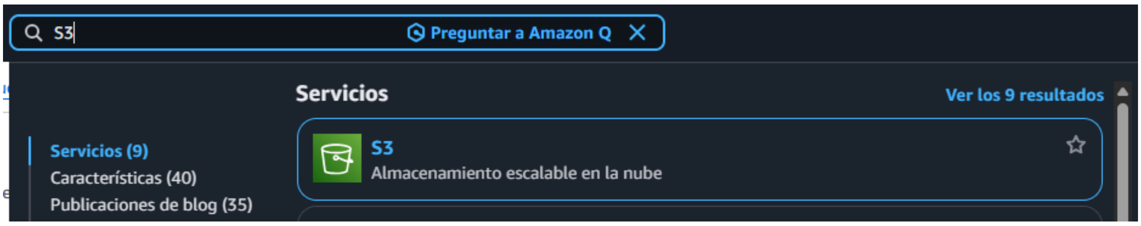
Por “by_station” y “by_year”



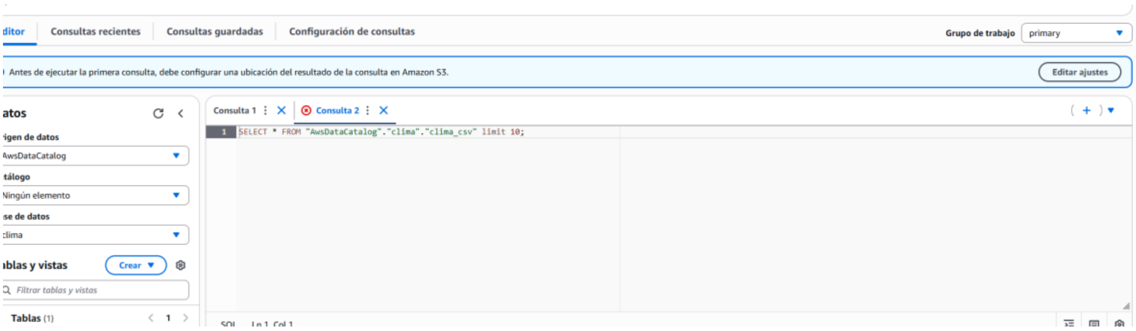
APARTADO C

Desde ATHENA, intenta realizar las siguientes consultas mostrando sus resultados y tiempos de ejecución:

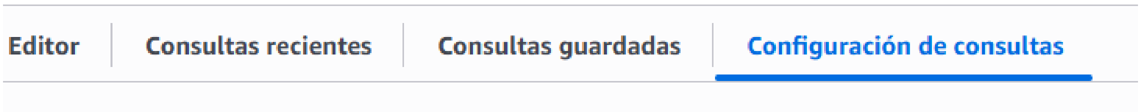
Primero configuramos el bucket para guardar las consultas

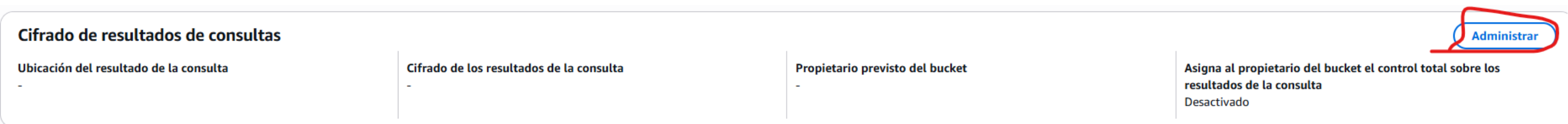


Después entramos a Athena desde “Table Data”

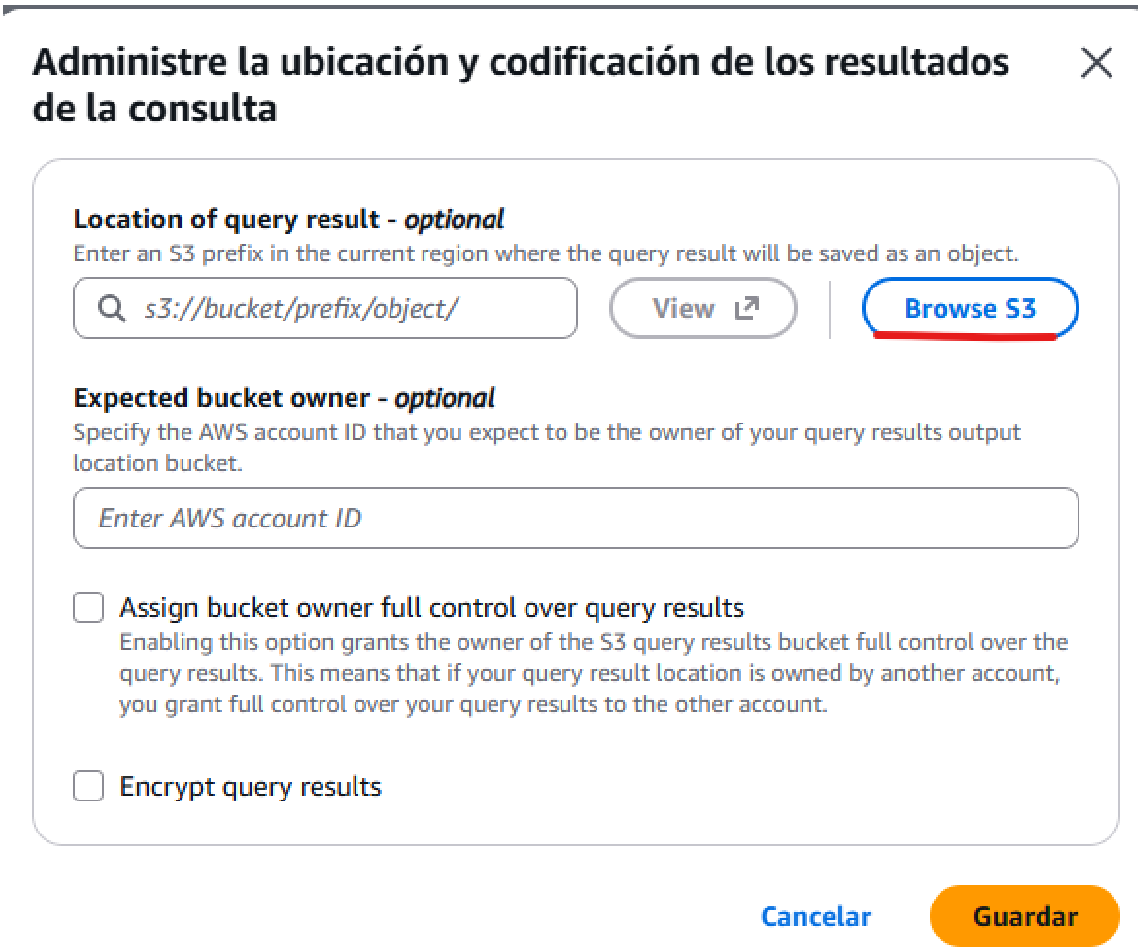


Para asignar donde se guardarán las consultas, entramos en Configuración de consultas y después en Administrar

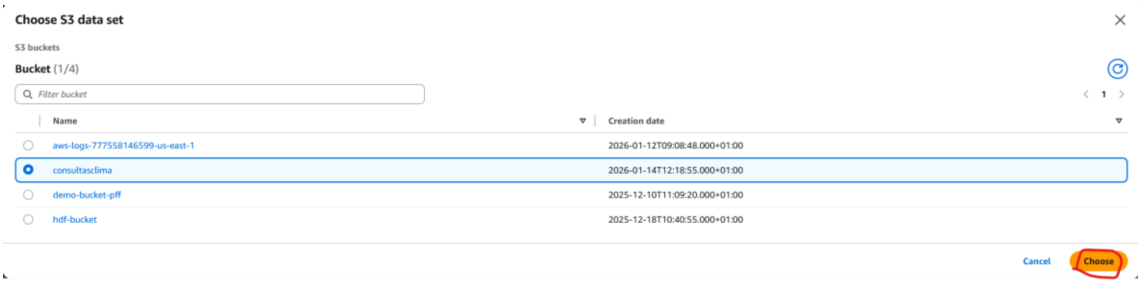




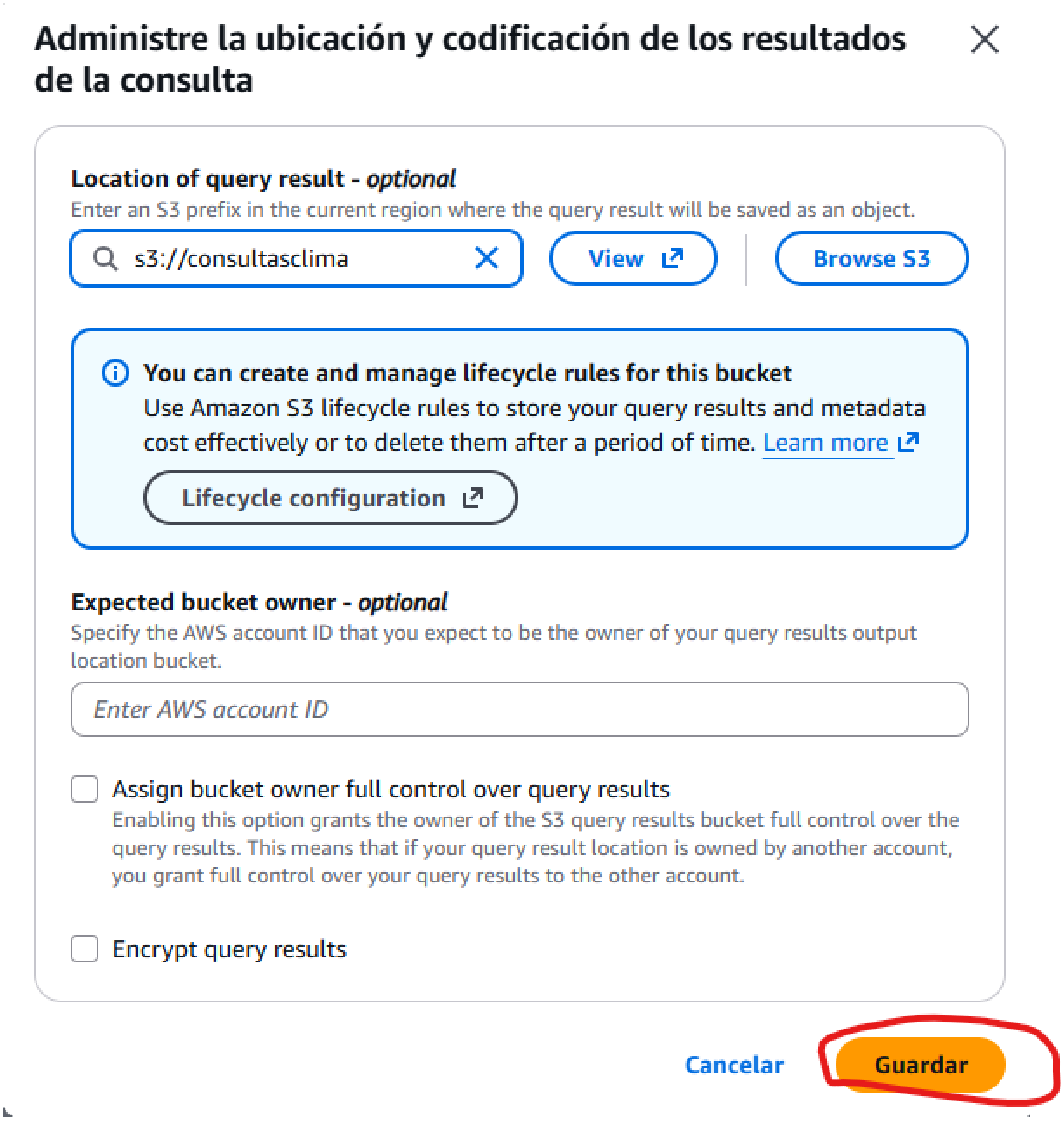
Ahora asignamos el bucket que hemos creado



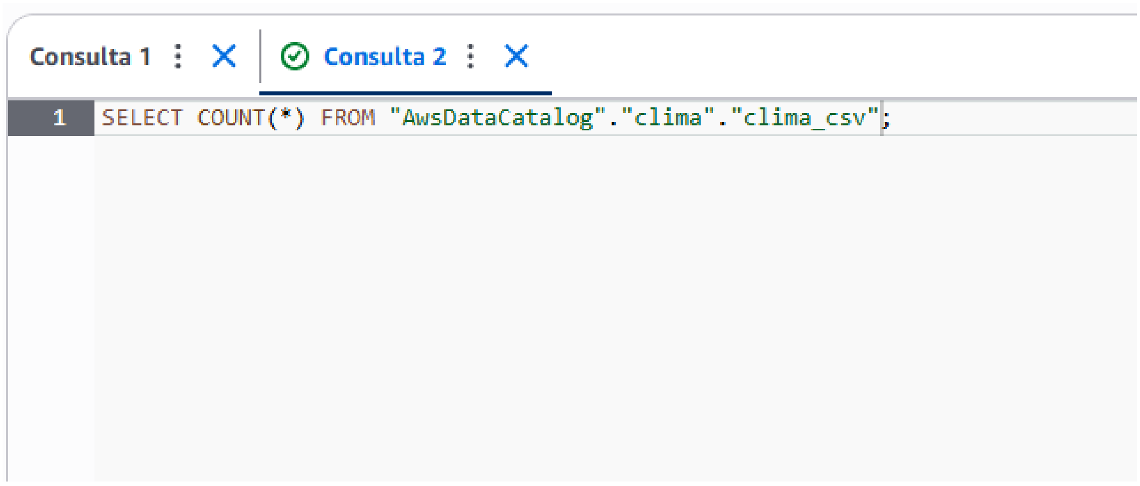
Administre la ubicación y codificación de los resultados de la consulta
Location of query result - optional
Enter an S3 prefix in the current region where the query result will be saved as an object.
s3://bucket/prefix/object/
View
Browse S3
Expected bucket owner - optional
Specify the AWS account ID that you expect to be the owner of your query results output location bucket.
Enter AWS account ID
Assign bucket owner full control over query results
Enabling this option grants the owner of the S3 query results bucket full control over the query results. This means that if your query result location is owned by another account, you grant full control over your query results to the other account.
Encrypt query results
Cancelar
Guardar



Choose S3 data set
S3 buckets
Bucket (1/4)
Filter bucket
1
Name
Creation date
aws-logs-777558146599-us-east-1
2026-01-12T09:08:48.000+01:00
consultasclima
2026-01-14T12:18:55.000+01:00
demo-bucket-pff
2025-12-10T11:09:20.000+01:00
hdf-bucket
2025-12-18T10:40:55.000+01:00
Cancel
Choose



1.- ¿Cuántos registros tiene la tabla?



Tiempo en cola: 104 ms **Tiempo de ejecución:** 12.092 sec **Datos analizados:** 206.68 GB

Resultados de la consulta | Estado de la consulta

Completado

Resultados (1)

Filas de búsqueda

#	_col0
1	6336799722

2.- ¿Cuántas mediciones tenemos de España?

Consulta 1 | Consulta 2 | Consulta 3

```
SELECT COUNT(*) FROM "clima"."clima_csv" where id like 'SP%';
```

Tiempo en cola: 2.863 sec **Tiempo de ejecución:** 16.501 sec **Datos analizados:** 206.68 GB

Resultados de la consulta | Estado de la consulta

Completado

Resultados (1)

Filas de búsqueda

#	_col0
1	21166382

3.- Sabiendo los códigos de las 4 estaciones de Asturias ¿Cuántas mediciones tenemos de Asturias?

Consulta 1 | Consulta 2 | Consulta 3 | Consulta 4

```
SELECT COUNT(*) FROM "clima"."clima_csv" where id like 'SPE00119792'
OR id like 'SPE00119801'
OR id like 'SPE00119819'
OR id like 'SPE00119828';
```

Tiempo en cola: 109 ms **Tiempo de ejecución:** 12.659 sec **Datos analizados:** 206.68 GB

Resultados de la consulta | Estado de la consulta

Completado

Resultados (1)

Filas de búsqueda

#	_col0
1	544046

4.- ¿Cuántas mediciones tenemos de Oviedo?

Consulta 5

```
SELECT COUNT(*) FROM "clima"."clima_csv" where id like 'SPE00119828';
```

Tiempo en cola: 132 ms **Tiempo de ejecución:** 30.021 sec **Datos analizados:** 206.68 GB

#	_col0
1	146094

5.- ¿Cuál es la medición más antigua de España, Asturias y Oviedo?

Consulta 5 | Consulta 6 | Consulta 11

```
(SELECT * FROM "clima"."clima_csv" WHERE id like 'SP%' ORDER BY date LIMIT 1)
UNION
(SELECT * from "clima"."clima_csv"  WHERE id like 'SPE00119792'
OR id like 'SPE00119801'
OR id like 'SPE00119819'
OR id like 'SPE00119828' ORDER BY date LIMIT 1)
UNION ALL
(SELECT * FROM "clima"."clima_csv" WHERE id like 'SPE00119828' ORDER BY date LIMIT 1)
```

Tiempo en cola: 109 ms **Tiempo de ejecución:** 43.033 sec **Datos analizados:** 620.04 GB

#	id	date	element	data_value	m_flag	q_flag	s_flag	obs_time	partition_0
1	SPE00119828	19721201	TMAX	130			E		by_year
2	SPE00119801	19381001	TMAX	192			E		by_year
3	SPE00155329	18961101	TMAX	155			E		by_year